



Fig. 27: Two frames of ILARTS imagery (left) (Image courtesy: wikipedia); the dash near the left side scale is the position of the aircraft with respect to glideslope. ILARTS control centre (right) (Image courtesy: US Navy)



Fig. 28: LSOs and LSO console showing an approaching F-18 aircraft (highlighted)

places on the flight-deck. These cameras videograph the approaching aircraft along with its relative position with the horizon. ILARTS then adds glideslope crosshairs parameters and other data to that imagery. The resultant composite video is displayed at the LSO console, CATCC and other important centres like CDC. The aircraft's alignment with respect to the horizon indicates the LSO whether the aircraft should climb up or descend so as to capture the glideslope of the pitching flight deck.

Fig. 27 indicates the two scenarios of an aircraft approaching for landing. The scale-like symbol on the left indicates acceptable limits for the aircraft in the vertical axis. The dash symbol near this scale indicates the position of the aircraft. The aircraft has to be ideally at the centre of the crosshairs. The dash indicates the actual position of the aircraft with respect to the glideslope. If the

aircraft is not within this envelope, it is waved off for another approach. These symbols are generated by ILARTS and fed to the LSO console.

AN/SPN-44. This is known as a range-rate radar. It detects approaching aircraft, tracks it and calculates its speed (both relative to the carrier and true speed). It then provides this data to the LSO console. Supplied with this accurate information on the speed of the approaching aircraft, LSOs can wave off the aircraft attempting to land at an unsafe speed.

Wind-measurement system. This is nothing but a traditional wind-measuring system based on an anemometer. It is used to measure headwind, tailwind and crosswind speeds. These winds can terribly affect the aircraft's flight by shifting the aircraft. Headwinds resist flight, tailwinds push the aircraft forward and crosswinds laterally move the aircraft out of glideslope. These winds are

troublesome. Data from the system is fed to the LSOs, who, in turn, intimate the same to the pilot. The pilot then flies in a way offsetting the effect of these winds.

Based on these inputs, LSOs communicate with the approaching pilot, who constantly adjusts aircraft's position based on the voice commands given by LSOs.

ACLS mode. Imagine that a pilot is returning to the carrier from a battlefield, which is located 500km away, after bombing enemy targets and evading enemy air defence missiles. On the way back, he or she had to intensely fight with enemy fighter aircraft. Now, on returning to the carrier, the pilot has to land his aircraft under pitch-black conditions. The pilot's physical state is not as alert as it was when he or she left for the mission; the intense adrenalin rush has tired the pilot, making landing of the aircraft even more difficult. Anyone in this situation would dream of a system that lands the aircraft automatically without any inputs. ACLS is that dream system.

Parts of this system exist in both the carrier and the aircraft. Only ACLS-equipped aircraft can be landed automatically on a carrier. Basically, in an ACLS, autopilot of the aircraft is controlled by the carrier. For this, position of the aircraft is tracked through Ka-band tracking radar AN/SPN-46 in the carrier. Instructions for the autopilot are transmitted to the aircraft. There are two radars guiding each aircraft.

The sequence starts from a point called acquisition window during approach of the aircraft. At this point, the tracking radar acquires the aircraft and starts tracking it. At this point the pilot is indicated through cockpit indicators that the aircraft is being acquired by ACLS. The pilot then puts the aircraft under autopilot in landing mode. The tracking radar uses a conical scan antenna and continuously interrogates a radar beacon present in the aircraft. This